A First Course in Causal Inference Ch. 9

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Ch. 9 Bridging Finite and Super Population Causal Inference

Definition 9.1 internal validity

The statistical analysis is valid for the study samples at hand.

Definition 9.2 external validity

The statistical analysis is valid for a broader population beyond the study samples.

- Do finite population results generalize to a bigger population?
- The external validity of the results depends on the sampling process of the units
  - If the finite population represents a sample of a larger population, the results also have external validity.
- The results based on randomization inference may not generalize; it ensures only internal validity.

Ch. 9 Bridging Finite and Super Population

To rigorously discuss this issue, we introduce a frmework with two levels of randomness:

- Randomness due to the sampling process of the units;
- Randomness due to the random assignment of the treatment.

Actually, this framework does not really solve the problem mentioned above, but it is beneficial for the following two purposes:

- it gives a different perspective on randomized experiments;
- it serves as a bridge between experimental studies and observational studies.

9.1 CRE

- Assume $\{Z_i, Y_i(1), Y_i(0), X_i\}_{i=1}^n \overset{\text{i.i.d.}}{\sim} \{Z, Y(1), Y(0), X\}$ from a superpopulation.
- With this assumption, we define the population average causal effect as $\tau = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$.

Definition 9.3 CRE under the superpopulation framework

We have

$$Z \perp \{Y(1), Y(0), X\}.$$

The average causal effect can be written as

$$\tau = \mathbb{E}[Y(1) \mid Z = 1] - \mathbb{E}[Y(0) \mid Z = 0] = \mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0].$$

9.1 CRE

- Explicitly, the first equation states that $\mathbb{E}[Y(1)] = \mathbb{E}[Y(1) \mid Z = 1]$ and it implies that the mean of potential outcomes and the potential outcomes in a treatment group is equivalent.
- This is the value of randomization: we can compare the *realized* outcomes across treatment and control groups.
- We have

$$\mathbb{E}[\hat{\tau} \mid Z] = \tau, \quad \text{var}[\hat{\tau} \mid Z] = \frac{\text{var}[Y(1)]}{n_1} + \frac{\text{var}[Y(0)]}{n_0}.$$

- Under i.i.d. sampling, the sample variances are unbiased for the population variances so Neyman (1923)'s variance estimator is unbiased for $\text{var}[\hat{\tau} \mid Z]$.

9.1 CRE

We can also discuss covariate adjustment. Based on the OLS decompositions,

$$Y(1) = \gamma_1 + \beta_1^T X + \varepsilon(1)$$
$$Y(0) = \gamma_0 + \beta_0^T X + \varepsilon(0),$$

we have

$$\tau = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = \gamma_1 - \gamma_0 + (\beta_1 - \beta_0)^T \mathbb{E}[X].$$

So the sample-version of the estimator is

$$\hat{\tau}_{\text{adj}} = \hat{\gamma}_1 - \hat{\gamma}_0 + (\hat{\beta}_1 - \hat{\beta}_0)^T \overline{X}.$$

Obviously, when $\overline{X} = 0$, $\hat{\tau}_{\text{adj}} = \hat{\tau}_L$ (Lin (2013)'s estimator.)

9.1 CRE

Taking difference between τ and $\hat{\tau}_{\text{adj}}$,

$$\tau - \hat{\tau}_{\text{adj}} = \underbrace{\{(\gamma_1 - \gamma_0) - (\hat{\gamma}_1 - \hat{\gamma}_0)\}\{\}}_{\text{Regression coefficient estimation error}} + \underbrace{\Delta\beta^T(\mathbb{E}[X] - \overline{X})}_{\text{uncertainy under the superpopulation framework}}$$

Hence, we introduce the proposed correction of the EHW estimator by adding an extra term

$$\frac{(\hat{\beta}_1 - \hat{\beta}_0)^T S_X (\hat{\beta}_1 - \hat{\beta}_0)}{n}.$$

9.3 Extention to the SRE

Definition 9.4 SRE under the superpopulation framework

We have

$$Z \perp \{Y(1), Y(0)\} \mid X.$$

The conditional average effect can be rewritten as

$$\tau_{[k]} = \mathbb{E}[Y(1) - Y(0) \mid X = k] = \mathbb{E}[Y \mid Z = 1, X = k] - \mathbb{E}[Y \mid Z = 0, X = k].$$

The average causal effect can be rewritten as

$$\tau = \mathbb{E}[Y(1) - Y(0)] = \sum_{k=1}^{K} \Pr(X = k)\tau_{[k]}.$$

References I

Lin, W. (2013). Agnostic notes on regression adjustments to experimental data: Reexamining freedman's critique. *The Annals of Applied Statistics*, 295–318.

Neyman, J. (1923). On the application of probability theory to agricultural experiments. essay on principles. *Ann. Agricultural Sciences*, 1–51.